

Akul Rajput

Senior Data Engineer | akulrajput1997@gmail.com | +91-XXXXXXXX

PROFESSIONAL SUMMARY

Senior Data Engineer with 6.9 years of experience in designing, developing, and optimizing scalable data pipelines, ETL/ELT workflows, and large-scale data processing solutions across Banking, Insurance, and Financial Services domains. Strong expertise in Python, SQL, PySpark, Apache Spark, Databricks, Apache Airflow, AWS, and Data Warehousing, with proven experience processing high-volume datasets and delivering reliable enterprise data solutions.

Experienced in building distributed data processing systems, developing complex SQL transformations, implementing data modeling techniques, and designing scalable data integration frameworks to support reporting, analytics, and business-critical decision-making. Proficient in workflow orchestration, data quality validation, performance optimization, and handling large-scale structured and semi-structured datasets.

Hands-on experience delivering end-to-end data engineering solutions, including ETL pipeline development, data reconciliation, data quality controls, process automation, and production support. Proven ability to collaborate with business stakeholders, mentor team members, drive technical solutions, and contribute to high-impact data initiatives across cross-functional teams.

PROFESSIONAL EXPERIENCE

Senior Data & Risk Analyst | Evaluateserve | Client: Global Investment Bank

Jan 2025– Present

- Designed and enhanced risk amendment and rollover frameworks supporting Product, Book, Factor Category, and Issue-level processing across enterprise risk platforms.
- Developed and optimized SQL-based data processing solutions using Stored Procedures, Dynamic SQL, Window Functions, and CTEs to support large-scale risk calculations and reporting workflows.
- Engineered scalable ETL and data transformation pipelines for Market Access Process (MAP) and Approved Product Registry (APR) platforms, improving processing efficiency and maintainability.
- Implemented complex business rules for MRE, DRC, and RRAO calculations, ensuring accurate risk factor rollovers across existing and newly onboarded positions.
- Developed Python and PySpark-based PnL processing pipelines for consolidation and analysis of large financial datasets consisting of millions of records.
- Performed data validation, reconciliation, and quality assurance activities across critical risk datasets, ensuring reporting accuracy and regulatory compliance.
- Collaborated with business analysts, risk teams, and technology stakeholders to deliver enhancements, production support, and process optimization initiatives.
- Contributed to automation and performance improvement initiatives, reducing manual effort and improving reliability of daily risk reporting processes.
- Designed and optimized scalable ETL/ELT pipelines and distributed data processing workflows for large-scale financial datasets, improving performance, reliability, and reporting efficiency.

Technical Lead | Niva Bupa |

Nov 2023 – Jan 2025

- Developed and maintained Python-based data processing solutions supporting premium calculation and financial reporting workflows.
- Leveraged Pandas and NumPy to perform large-scale data cleansing, validation, and transformation, ensuring high-quality datasets for downstream reporting and analytics.
- Designed and implemented optimized business logic for premium calculations, reducing processing time by 40% and improving operational efficiency.
- Performed data validation, reconciliation, and quality checks across financial datasets to ensure reporting accuracy and data integrity.
- Collaborated with cross-functional business and technology teams to gather requirements and deliver scalable data-driven solutions.
- Automated reporting workflows and managed end-to-end reliability of premium reporting processes used by finance and senior management teams.
- Worked closely with actuarial, finance, and business teams to validate premium and claims datasets before management and regulatory reporting.
- Led requirement discussions with stakeholders and translated business requirements into scalable reporting and data processing solutions.

Senior Software Developer | Newgen Software Technologies |

Jul 2021 – Nov 2023

- Analyzed and optimized complex Python-based processing workflows, resolving critical issues and improving overall system performance.
- Enhanced document scanning and data extraction processes, increasing throughput and improving processing accuracy for large-scale document workloads.
- Collaborated on multiple enterprise projects to integrate new features and improve interoperability across existing systems.
- Designed and prototyped scalable solutions for system integration, contributing to improved application performance and maintainability.
- Conducted code reviews and mentored junior developers, ensuring adherence to coding standards and software quality best practices.

Assistant Software Engineer | Tata Consultancy Services

Jan 2020 – Jul 2021

- Developed Python-based data extraction solutions using Tesseract OCR and OpenCV to process large volumes of medical records and scanned documents.
- Led the Pfizer project, designing and implementing automated document-processing workflows that improved extraction accuracy and operational efficiency.
- Built scalable OCR and data-processing systems for large-scale document ingestion, extraction, and validation.
- Collaborated with cross-functional teams to deliver high-quality solutions, ensuring timely project execution and business alignment.
- Improved text extraction accuracy and processing speed through optimization of OCR pipelines and custom image-processing techniques.

Graduate Engineering Trainee | Credicxo

Jul 2019 – Sep 2019

- Developed and enhanced bulk data-processing features, improving application performance and runtime efficiency through multi-processing techniques.
- Optimized data scraping and image-classification workflows, increasing processing accuracy and overall system effectiveness.
- Contributed to the development of a loan-lending platform by implementing solutions for automated verification of pre-approved loan agreements.
- Integrated multi-processing modules to improve large-scale data ingestion and processing from enterprise repositories.

TECHNICAL SKILLS SNAPSHOT

- **Languages:** Python, SQL
- **Big Data & Data Processing:** PySpark, Apache Spark, Pandas, NumPy, Distributed Data Processing, Large-Scale Data Processing
- **Data Platforms:** Databricks
- **Data Engineering:** ETL, ELT, Data Warehousing, Data Modeling, Star Schema, Snowflake Schema, SCD Type 1 & Type 2, Data Integration
- **Databases:** MySQL, SQL Server
- **Workflow & Orchestration:** Apache Airflow, Workflow Orchestration
- **Cloud & DevOps:** AWS, Docker, Git, GitHub
- **Analytics & Reporting:** Data Validation, Data Quality, Financial Reporting, Risk Analytics, Data Reconciliation
- **Computer Vision & OCR:** OpenCV, Tesseract OCR
- **Machine Learning:** Scikit-Learn

Education

- Bachelor of Technology (Computer Science & Engineering) RD Engineering College (AKTU)